

An Introduction to The String Data Type

A string is a sequence of characters such as names, addresses, car plates or meaningless sequence of characters such as “h83hdu2&e\$8”. String is not a primitive data type. Rather, String is an object type. The string always ends with ‘\0’ or null character. In Java, we could declare a string the same way as we declare variables:

```
String name;
```

We could assign this string a value:

```
name="java";
```

Note that the string index begins at 0.

index	0	1	2	3	4
name	j	a	v	a	\0

We could obtain this string from the keyboard using next() or nextLine() method of Scanner:

```
name = next();
```

```
name = nextLine();
```

The following table lists a number of interesting and useful methods of the string:

Method Name	Description	Output
length	Returns the length of the string.	int
charAt(i)	Returns the character at that position (i) in the string.	char
Substring(i,j)	Returns some characters from i to j-1 in the string.	String
concat	Accepts a string and add it to the end of the original string.	String
toUpperCase	Returns a copy of the original string, all upper case.	String
toLowerCase	Returns a copy of the original string, all lower case.	String
compareTo	Compares two strings. It returns 0 if the strings are identical, a negative number if the first string comes before the second in the alphabet, and a positive number if it comes later.	int
equals	Compares two strings. It returns true if these are identical, otherwise returns false .	boolean
equalsIgnoreCase	Compares two strings. It returns true if these are identical (ignoring case), otherwise returns false .	boolean
startsWith	Accepts a string and returns true if the original string starts with this string and false if it does not.	boolean
endsWith	Accepts a string and returns true if the original string ends with this string and false if it does not.	boolean
trim	Removes any spaces at the beginning or end of a string.	String

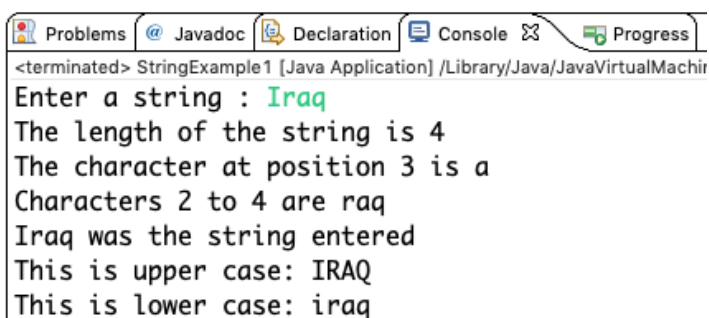
For examples,

```
String str = "Java is cool!";
System.out.println(str.length());           // return int 13
System.out.println(str.charAt(2));           // return char 'v'
System.out.println(str.charAt(5));           // return char 'i'
System.out.println(str.substring(0, 3));     // return String "Jav"
System.out.println(str.endsWith("cool!"));  // return boolean true
str = "    Java is cool!    ";
System.out.println(str.trim());              // return String "Java is cool!"

// Comparing two Strings
String str = "Kufa university";
String anotherStr = "Kufa UNIVERSITY";
System.out.println(str.equals(anotherStr));  // return boolean false
System.out.println(str.equalsIgnoreCase(anotherStr)); // return boolean true
System.out.println(anotherStr.equals(str));  // return boolean false
System.out.println(anotherStr.equalsIgnoreCase(str)); // return boolean true

import java.util.Scanner;
public class StringExample1 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        String str;
        System.out.print("Enter a string : ");
        str = input.next();
        input.close();

        System.out.println("The length of the string is " + str.length());
        System.out.println("The character at position 3 is " + str.charAt(2));
        System.out.println("Characters 2 to 4 are " + str.substring(1,4));
        System.out.println(str.concat(" was the string entered"));
        System.out.println("This is upper case: " + str.toUpperCase());
        System.out.println("This is lower case: " + str.toLowerCase());
    }
}
```



```
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<terminated> StringExample1 [Java Application] /Library/Java/JavaVirtualMachin
Enter a string : Iraq
The length of the string is 4
The character at position 3 is a
Characters 2 to 4 are raq
Iraq was the string entered
This is upper case: IRAQ
This is lower case: iraq
```

```

import java.util.Scanner;
public class StringExample2 {
    public static void main(String[] args) {
        String string1, string2;
        int result;
        Scanner keyboard = new Scanner(System.in);
        System.out.print("Enter a String: ");
        string1 = keyboard.nextLine();
        System.out.print("Enter another String: ");
        string2 = keyboard.nextLine();
        keyboard.close();

        result = string1.compareTo(string2);
        if(result < 0)
            System.out.println(string1 + " comes before " + string2 + " in the
alphabet");
        else if(result > 0)
            System.out.println(string2 + " comes before " + string1 + " in the
alphabet");
        else
            System.out.println("The strings are identical");
        }
    }
}

```

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<terminated> StringExample2 [Java Application] /Library/Java/JavaVirtualMachir

Enter a String: Kufa

Enter another String: Kufa

The strings are identical

Problems Javadoc Declaration Console Progress

<terminated> StringExample2 [Java Application] /Library/Java/JavaVirtualMachir

Enter a String: ali

Enter another String: ahmed

ahmed comes before ali in the alphabet

Problems Javadoc Declaration Console Progress

<terminated> StringExample2 [Java Application] /Library/Java/JavaVirtualMachir

Enter a String: Najaf

Enter another String: najaf

Najaf comes before najaf in the alphabet